

Problem Statement ID - 26125

Problem Statement Title - Blockchain-Based Secure Platform for Identity, Access Control, and Digital Asset Management

- Theme** - Blockchain & Cybersecurity
- PS Category** - Software
- Organization** - Bharat Electronics Limited (BEL)
- Team ID** - SIH2026-BEL-7492
- Team Name** - TRUSTFORGE INNOVATORS



A SECURE BLOCKCHAIN ECOSYSTEM FOR IDENTITY, ACCESS CONTROL & ASSET MANAGEMENT

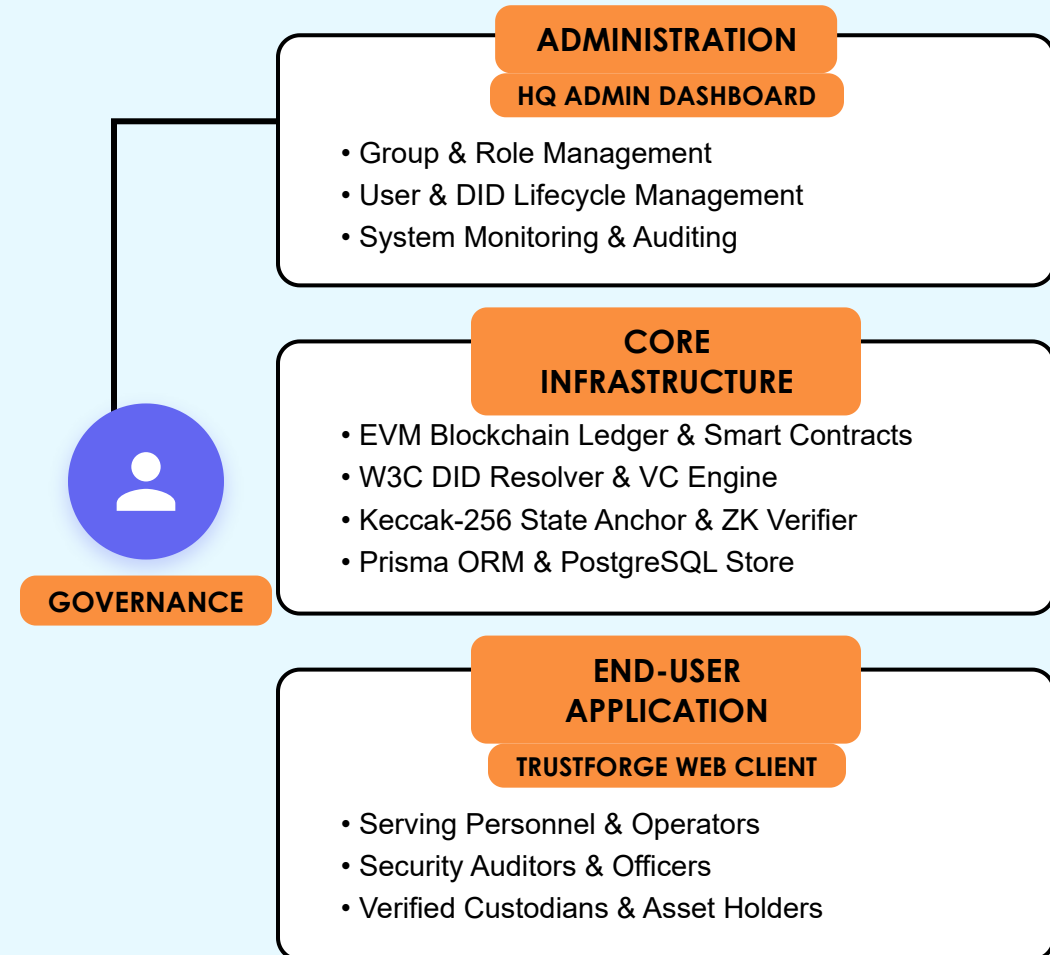
PROBLEM OVERVIEW

- Defence and enterprise personnel relying on centralized IAM systems risk catastrophic data leaks and single points of failure.
- Foreign-hosted IAM platforms expose sensitive defence infrastructure to cyber espionage and unauthorized access.
- No sovereign, tamper-proof blockchain system exists for unified identity, RBAC, and digital asset tracking.
- Commercial IAM apps lack strict cryptographic access control, verifiable provenance, and data breach prevention.
- Sensitive operational records and asset custody history remain vulnerable to manipulation on centralized databases.

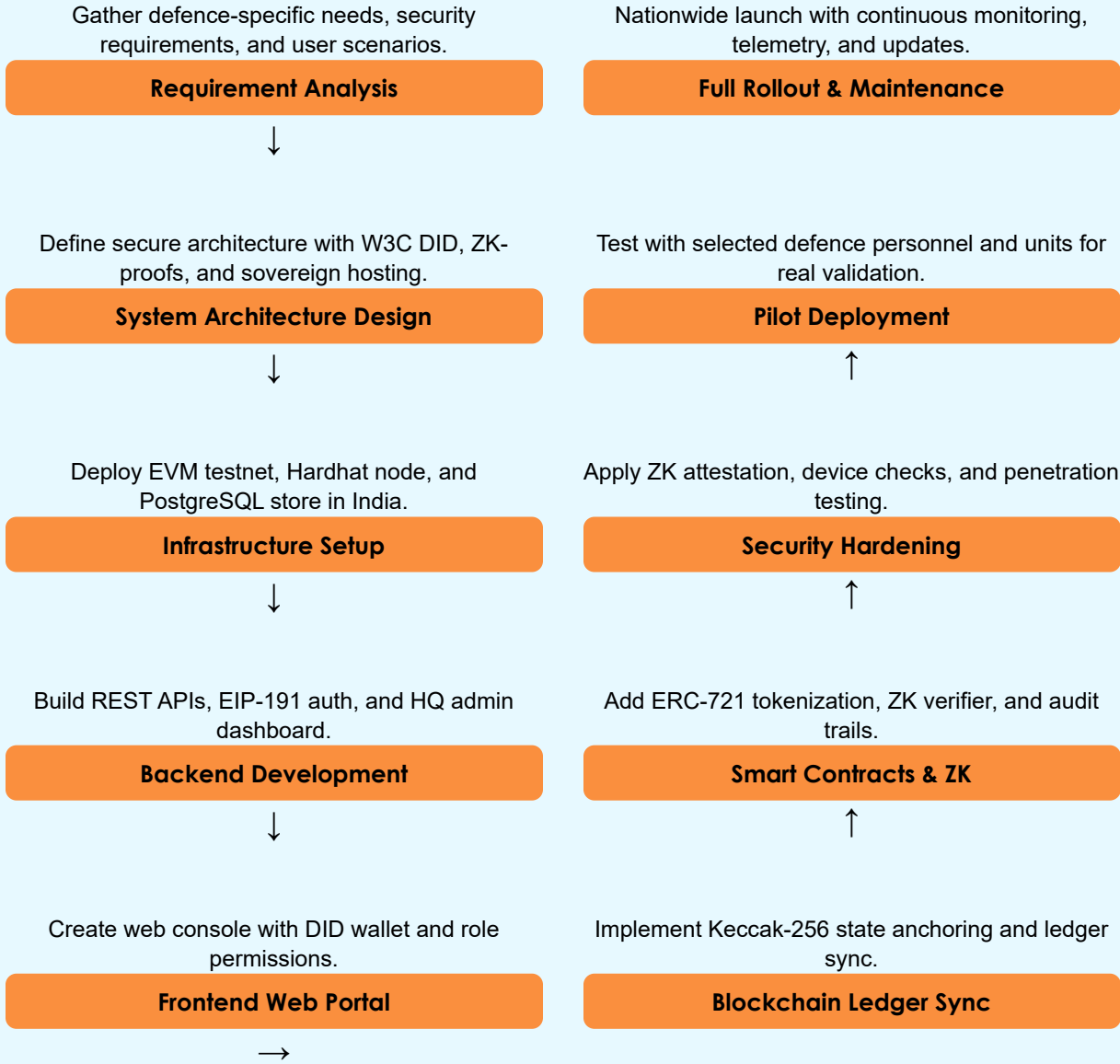
PROBLEM SOLUTION

- **W3C-Verified DIDs** – Only verified personnel, officers, and registered devices hold cryptographic self-sovereign DIDs.
- **ERC-721 Tokenized Assets** – Physical and digital equipment minted as non-fungible tokens directly tied to owner DIDs.
- **Smart Contract RBAC** – Role-based access control deterministically governed on-chain without central vulnerability.
- **Keccak-256 State Anchoring** – All access events and credential hashes permanently locked on-chain for tamper-proof auditing.
- **On-Premise Sovereign Hosting** – Fully hosted within Indian territory ensuring 100% data sovereignty under defence oversight.

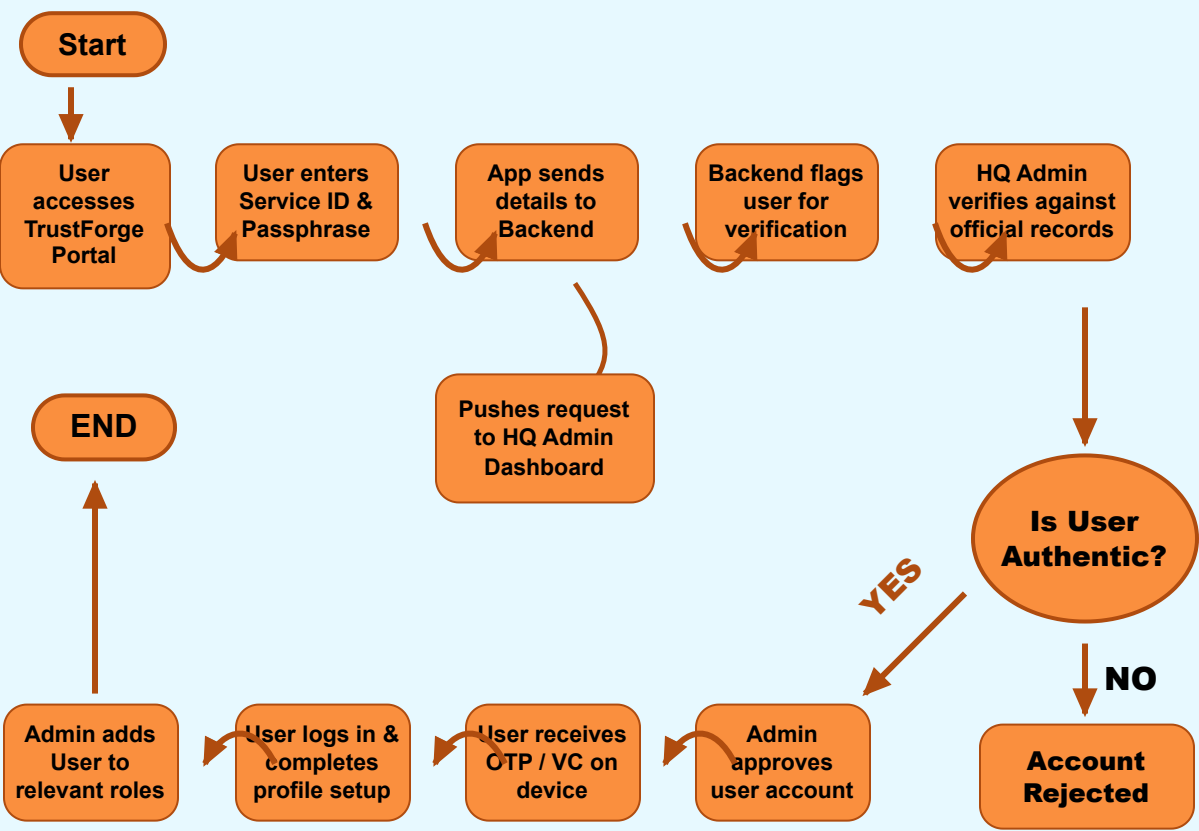
HIERARCHICAL DIAGRAM



METHODOLOGY



FLOW CHART:



ANALYSIS OF THE FEASIBILITY OF THE IDEA

1. TECHNICAL FEASIBILITY	2. ECONOMICAL FEASIBILITY	3. SOCIAL FEASIBILITY	4. LEGAL & COMPLIANCE FEASIBILITY	5. OPERATIONAL FEASIBILITY	6. SECURITY FEASIBILITY
Robust EVM blockchain, W3C DIDs, ZK proofs, and on-premise infrastructure are achievable using mature open-source stacks and hardened backend engines.	While initial setup requires infrastructure investment, long-term operational costs remain minimal through self-hosted EVM nodes, zero licensing fees, and microservices.	The platform builds deep trust among defence units and enterprise teams by guaranteeing sovereign identity ownership and tamper-proof operational record protection.	All data is strictly hosted within Indian sovereign territory, ensuring full compliance with CERT-In directives, DPDP Act 2023, and defence cybersecurity frameworks.	An intuitive web interface with role-based dashboards ensures frictionless onboarding, while centralized HQ-admin controls streamline lifecycle verification and audits.	Continuous penetration testing, DevSecOps pipelines, cryptographic smart contract audits, and zero-trust perimeter defense ensure defense against state-sponsored actors.

POTENTIAL CHALLENGES AND RISKS

SECURITY VULNERABILITIES

Risk of cyberattacks from state-sponsored actors targeting smart contracts or node RPC endpoints.



USER ADOPTION & TRAINING

Personnel and enterprise operators may resist shifting from traditional legacy IAM platforms.



PERFORMANCE & SCALABILITY

Ensuring ultra-low latency transaction throughput under high concurrent load across distributed military commands.



INSIDER THREATS

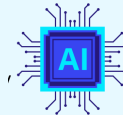
Malicious administrative actions or unauthorized credential escalation from legitimate privileged users.



STRATEGIES FOR OVERCOMING THESE CHALLENGES

SECURITY HARDENING

Formal smart contract verification, automated DevSecOps pipelines, HSM keystores, and regular third-party audits.



AWARENESS & TRAINING

Zero-learning-curve UI, interactive training modules, and transparent onboarding explaining cryptographic benefits.



OPTIMIZED ARCHITECTURE

Layer-2 rollup integration, batch state commitments, Redis caching, and offline-first edge cryptographic verification.



ACCESS CONTROL & MONITORING

Enforce strict least-privilege RBAC, multi-signature admin thresholds, and tamper-proof immutable audit logging.



IMPACT AND BENEFITS

UNIQUE FEATURES:

Mandatory DID Verification

All identity claims are anchored cryptographically to W3C-compliant standards.

Smart Contract RBAC

Access permissions executed deterministically on-chain without central points of failure.

ERC-721 Asset Custody

Physical and digital military assets tracked with immutable lifecycle provenance.

Zero-Knowledge Attestation

Selective disclosure of clearances without exposing sensitive underlying PII data.

Sovereign Data Hosting

100% infrastructure self-hosted within Indian borders under national oversight.

IMPACTS

Enhanced National Security



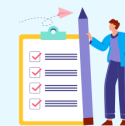
Eliminates single point of failure in critical defence IAM networks.

Improved OPSEC



Prevents hostile intelligence exploitation of identity credentials and locations.

Asset Integrity Protection



Guarantees full provenance and chain of custody for tokenized assets.

Digital Sovereignty



Eliminates reliance on foreign IAM providers, ensuring true technological independence.

Resilient Defence Infrastructure



Builds decentralized zero-trust framework across tri-service commands.

BENEFITS

Trustworthy Ecosystem



Provides a mathematically verifiable security realm for all personnel.

High Enterprise Morale



Boosts confidence knowing sensitive personnel data cannot be leaked.

Efficient Auditing



Automated on-chain event verification cuts manual audit time by 90%.

Scalable Solution



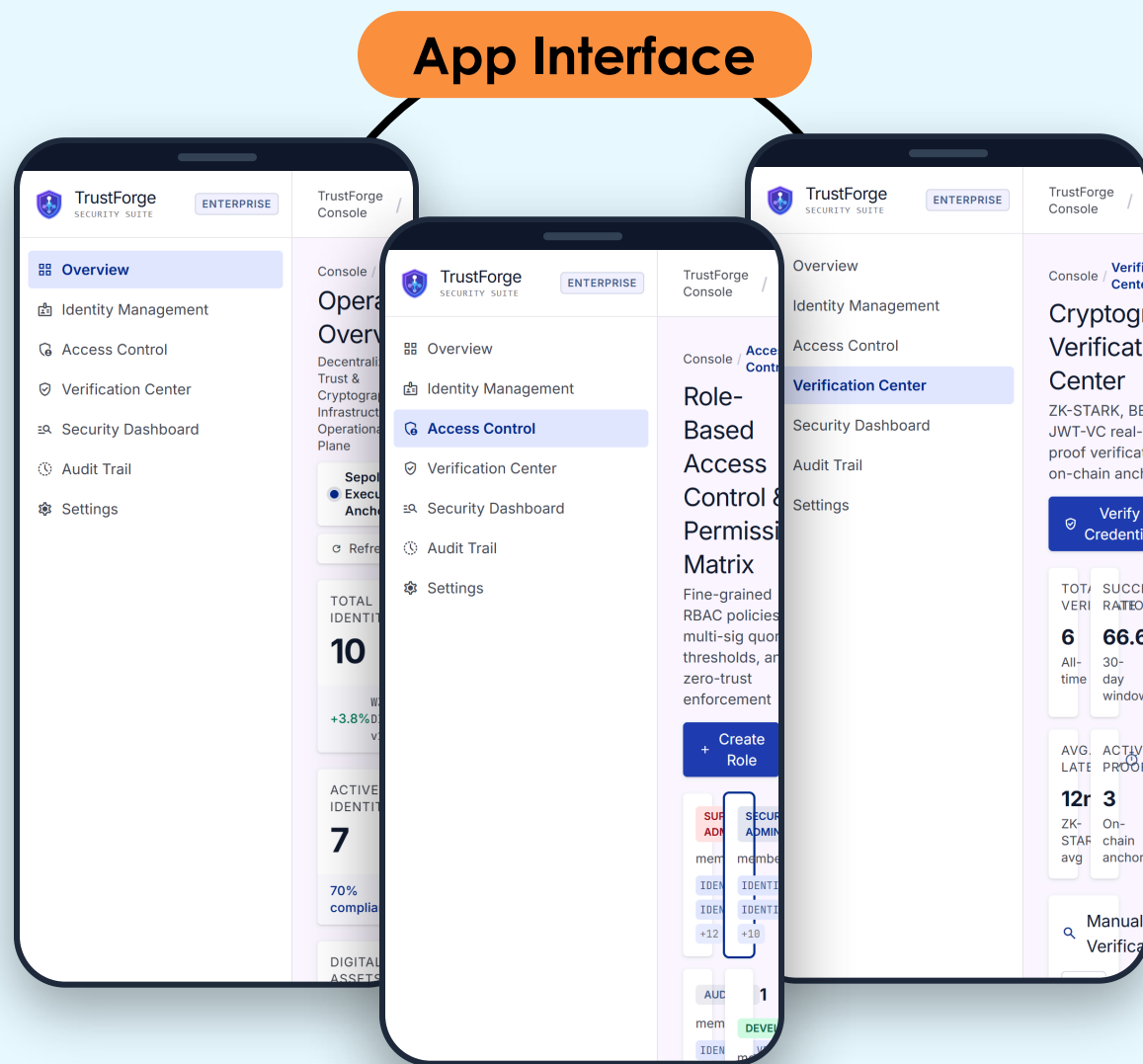
Modular EVM smart contracts scale seamlessly to millions of records.

Long-Term Cost Value



Permanent open-source platform eliminates recurrent vendor software fees.

OUR PROTOTYPE



Web & Mobile Interface

RESEARCH AND REFERENCES

• W3C Decentralized Identifiers (DIDs) v1.0

Summary: Defines verifiable, globally unique, cryptographically sovereign identifiers operating independent of centralized registry authorities.

Link: <https://www.w3.org/TR/did-core/>

• Ethereum ERC-721 Non-Fungible Token Standard

Summary: Core standard for unique, indivisible on-chain tokens utilized to represent physical equipment custody and verifiable asset provenance.

Link: <https://eips.ethereum.org/EIPS/eip-721>

• Zero-Knowledge Cryptographic Proof Systems

Summary: Mathematical verification proofs enabling selective clearance validation without revealing sensitive identities, clearances, or underlying PII.

Link: <https://zkproof.org/standard/>

• CERT-In Cybersecurity Guidelines for Critical Infra

Summary: Sovereign cybersecurity compliance protocols enforcing immutable audit logs, strict role boundaries, and domestic data localization.

Link: <https://www.cert-in.org.in/guidelines/>